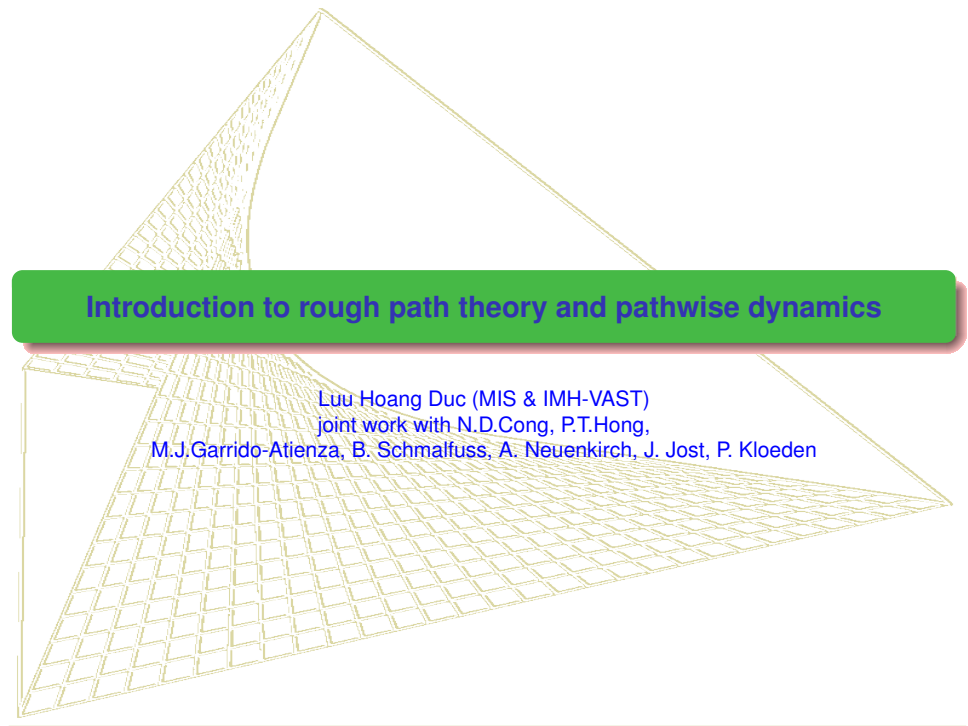




Introduction to rough path theory and pathwise dynamics

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History

- L.C. Young (1936). An integration of Hölder type, connected with Stieltjes integration. Acta Math.
- K. T. Chen (1954). "Iterated Integrals and Exponential Homomorphisms". Proceedings of the London Mathematical Society.
- H. J. Sussmann (1978). On the gap between deterministic and stochastic ordinary differential equations. Annals of Prob. **pathwise approach for solving Stratonovich SDE**
- T. Lyons (1998). Differential equations driven by rough signals. Revista Matematica Iberoamericana. **Universal limit theorem: birth of Rough path theory!**
- D. Nualart, M. Hairer, M. Gubinelli, P. Friz, L. Coutin,
- Dynamical approach: M. Scheutzow, B. Schmalfuss, LHD...
- Since 2014, after Hairer's Fields medal: more applications in data analysis, ML, time series analysis.
- Truncated tensor algebra $T^{(n)}(\mathbb{R}^d) = \bigoplus_{i=0}^n (\mathbb{R}^d)^{\otimes i}$.
- Geometric rough paths: $\mathbf{X}_{s,t} = \mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t}$ for all $s \leq u \leq t$ where

$$\mathbf{X}_{s,t} = \left(1, \int_{s \leq s_1 \leq t} dX_{s_1}, \dots, \int_{s \leq s_1 \leq \dots \leq s_{[p]} \leq t} dX_{s_1} \otimes \dots \otimes dX_{s_{[p]}} \right)$$

- Universal theorem (T. Lyons 1998): convergence of smooth paths to rough paths and continuity of solution w.r.t. driving paths.

(controlled) Rough differential equations

$$dy = g(y)dx, \quad g \in C_b^3, y \in \mathbb{R}^d. \quad (1.1)$$

- **Lyons-Davie** solution: $y \in C^{p-\text{var}}([0, T], \mathbb{R}^d)$ is a solution if

$$\|y_{s,t} - g(y_s)x_{s,t} - Dg(y_s)g(y_s)\mathbb{X}_{s,t}\| \leq C(t-s)^{1+\epsilon}, \quad \forall 0 \leq s \leq t \leq T.$$

- **Friz-Victoir** solution: $y \in C^{p-\text{var}}([0, T], \mathbb{R}^d)$ is a solution if there exists a sequence $x^n \in C^\infty([0, T])$ such that $x^n \rightarrow x$ in $C^{p-\text{var}}$ with $\sup_n \|x^n\|_{p-\text{var}, [0, T]} < \infty$ and the solution

$$\dot{y}_t^n = g(y_t^n)\dot{x}_t^n$$

satisfies $y^n \rightarrow y$ in $C^{p-\text{var}}([0, T])$.

- **Gubinelli** solution: interpret (1.4) in the integration form

$$y_t = y_0 + \int_0^t g(y_s)dx_s$$

where the integral is understood as a rough integral.

Norms and spaces

- For $p \geq 1$, $C^{p-\text{var}}([a, b], H)$: the Banach space with finite norm

$$\|u\|_{p,[a,b]} = \|u_a\| + \|u\|_{p,[a,b]} = \|u_a\| + \|u\|_{p,[a,b]} = \left(\sup_{\Pi(a,b)} \sum_{[s,t] \in \Pi} |u_{s,t}|^p \right)^{1/p} < \infty.$$

- For $\beta \in (0, 1)$, $C^{\beta-\text{Hol}}([a, b], H)$ the Banach space equipped with norm

$$\|u\|_{\infty,\beta,[a,b]} := \|u\|_{\infty,[a,b]} + \|u\|_{\beta,[a,b]} = \|u\|_{\infty,[a,b]} + \sup_{a \leq s < t \leq b} \frac{\|u(t) - u(s)\|}{(t-s)^\beta}.$$

- For $\alpha, \beta \in (0, 1)$, $C^{\alpha,\beta}([a_+, b], H)$, $C^{\alpha,\beta}([a, b_-], H)$ the Banach spaces equipped with norms

$$\|u\|_{\infty,\alpha,\beta,[a_+,b]} := \|u\|_{\infty,[a,b]} + \sup_{a < s < t \leq b} |s-a|^\alpha \frac{\|u(t) - u(s)\|}{(t-s)^\beta}$$

$$\|u\|_{\infty,\alpha,\beta,[a,b_-]} := \|u\|_{\infty,[a,b]} + \sup_{a \leq s < t < b} |b-t|^\alpha \frac{\|u(t) - u(s)\|}{(t-s)^\beta}.$$

- One can prove that $\|y\|_{\alpha,\beta,[a_+,b]} = \sup_{a < s < b} (s-a)^\alpha \|y\|_{\beta,[s,b]}$, thus $C^\beta([a, b]) \subset C^{\beta,\beta}([a_+, b]) \cap C^{\beta,\beta}([a, b_-])$ with $\|u\|_{\beta,\beta,[a_+,b]} \leq |b-a|^\beta \|u\|_{\beta,[a,b]}$.

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^\beta([a, b], H)$ and $\mathbb{X} \in C_2^{2\beta}([a, b]^2, H \otimes H)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b. \quad (1.2)$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$. In a general form, $(y \otimes x)_{s,t} =: \int_s^t y_{s,u} \otimes dx_u$ satisfies Chen's relation

$$(y \otimes x)_{s,t} - (y \otimes x)_{s,u} - (y \otimes x)_{u,t} = y_{s,u} \otimes x_{u,t}.$$

Denote by $\mathcal{C}^\beta([a, b]) \subset C^\beta \oplus C^{2\beta}$ the set of all rough paths in $[a, b]$, then $\mathcal{C}^\beta$ is a closed set but not a linear space, equipped with the rough path semi-norm

$$\|\mathbf{x}\|_{\beta, [a, b]} := \|x\|_{\beta, [a, b]} + \|\mathbb{X}\|_{2\beta, [a, b]}^{\frac{1}{2}} < \infty. \quad (1.3)$$

Examples in 1D

- $X_t = \int_0^t a_s dB_s \Rightarrow \mathbb{X}_{s,t} := \int_s^t X_{s,u} dX_u = \frac{1}{2} X_{s,t}^2 - \frac{1}{2} \int_s^t a_u^2 du$. Integral in the Itô sense.
- $X_t = B^H \Rightarrow \mathbb{X}_{s,t} := \int_s^t B_{s,u}^H \delta B_u^H = \frac{1}{2} (B_{s,t}^H)^2 - \frac{1}{2} (t^{2H} - s^{2H})$. Integral in the Skorohod-Wick-Itô sense.
- $[x, 2]_{s,t} := x_{s,t}^2 - 2\mathbb{X}_{s,t} \Rightarrow [x, 2]_{s,t} = [x, 2]_{s,u} + [x, 2]_{u,t} \Rightarrow [x, 2]_{0,\cdot} \in C^{2\nu}$ is a Hölder continuous path.

$\beta > \frac{1}{2}$: Young integrals

- Consider $y \in C^\beta([a, b], V \otimes H)$ and $x \in C^\beta([a, b], H)$, there exists a Young integral $\int_a^b y dx$ defined by

$$\int_a^b y dx = \lim_{|\Pi| \rightarrow 0} \sum_{[s,t] \in \Pi} y_s x_{s,t},$$

- Young-Loeve (Y-L) estimate

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} \right\| \leq K_\beta |t - s|^{2\beta} \|y\|_{\beta, [a,b]} \|x\|_{\beta, [a,b]}.$$

In fact, we can prove using fractional calculus that

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} \right\| \leq K_p \|y\|_{p, [s,t]} \|x\|_{p, [s,t]};$$

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} \right\| \leq K_\beta \left(\|y_{s, \cdot}\|_{\infty, [s,t]} + \|y\|_{\beta, \beta, [s, t_+]} \right) \left(\|x_{\cdot, t}\|_{\infty, [s,t]} + \|x\|_{\beta, \beta, [s, t_-]} \right).$$

$\beta \in (\frac{1}{3}, \frac{1}{2})$: Rough integrals

- A path $y \in C^\beta([a, b], V \otimes H)$ is then called to be *controlled by* $x \in C^\beta([a, b], H)$ if there exists a tube (y', R^y) with $y' \in C^\beta([a, b], V \otimes H \otimes H)$, $R^y \in C^{2\beta}(\Delta^2([a, b]), V \otimes H)$ such that

$$y_{s,t} = y'_s x_{s,t} + R^y_{s,t}, \quad \forall \min I \leq s \leq t \leq \max I.$$

y' is called **Gubinelli derivative** of y , which is uniquely defined as long as $x \in C^\beta \setminus C^{2\beta}$ is truly rough (works for fBm B^H , $H \in (\frac{1}{3}, \frac{1}{2}]$ FH[Chapter 6]).

- **Sewing lemma**: denote $F_{s,t} := y_s x_{s,t} + y'_s \mathbb{X}_{s,t}$. Then

$$\begin{aligned} \delta F_{s,u,t} &:= F_{s,t} - F_{s,u} - F_{u,t} = -y'_{s,u} \mathbb{X}_{u,t} - R^y_{s,u} x_{u,t} \\ \Rightarrow \quad \|\delta F_{s,u,t}\| &\leq |t-s|^{3\beta} \left(\|x\|_{\beta,[s,t]} \|R^y\|_{2\beta,\Delta^2[s,t]} + \|y'\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \right) \end{aligned}$$

Thanks to the sewing lemma, the rough integral $\int_s^t y_u dx_u$ can be defined as

$$\int_s^t y_u dx_u := \lim_{|\Pi| \rightarrow 0} \sum_{[u,v] \in \Pi} [y_u x_{u,v} + y'_u \mathbb{X}_{u,v}]$$

Moreover, one gets Y-L typed estimate:

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} - y'_s \mathbb{X}_{s,t} \right\| \leq C_\beta |t-s|^{3\beta} \left(\|x\|_{\beta,[s,t]} \|R^y\|_{2\beta,\Delta^2[s,t]} + \|y'\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \right).$$

Existence and uniqueness theorem: RDEs

$$dy = g(y)dx, \quad g \in C_b^3, y \in \mathbb{R}^d. \quad (1.4)$$

- **Lyons-Davie** solution: $y \in C^{p-\text{var}}([0, T], \mathbb{R}^d)$ is a solution if

$$\|y_{s,t} - g(y_s)x_{s,t} - Dg(y_s)g(y_s)\mathbb{X}_{s,t}\| \leq C(t-s)^{1+\epsilon}, \quad \forall 0 \leq s \leq t \leq T.$$

- **Friz-Victoir** solution: $y \in C^{p-\text{var}}([0, T], \mathbb{R}^d)$ is a solution if there exists a sequence $x^n \in C^\infty([0, T])$ such that $x^n \rightarrow x$ in $C^{p-\text{var}}$ with $\sup_n \|x^n\|_{p-\text{var}, [0, T]} < \infty$ and the solution

$$\dot{y}_t^n = g(y_t^n)\dot{x}_t^n$$

satisfies $y^n \rightarrow y$ in $C^{p-\text{var}}([0, T])$.

- **Gubinelli** solution: interpret (1.4) in the integration form

$$y_t = y_0 + \int_0^t g(y_s)dx_s$$

where the integral is understood as a rough integral.

Existence and uniqueness theorem: mixed RDEs

- Finite dimension: RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ is understood in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)dx_s. \quad (1.5)$$

- Riedel and Scheutzow (2017)**: f is one sided Lipschitz and linear growth in perpendicular direction. $g \in C_b^3$ as usual. **LHD (2021)**: solution of the form (y, y') with $y, y' \in C^\beta$, $y' = g(y)$, which is solved by Doss-Sussmann technique.
- Infinite dimension - **M.J. G-A, LHD, B.Sch., A.N. - (2018)**: consider Young evolution equation

$$dy_t = [Ay_t + f(y_t)]dt + g(y_t)dx_t, \quad (1.6)$$

with the solution understood in the mild form

$$y_t = S(t)y_0 + \int_0^t S(t-s)f(y_s)ds + \int_0^t S(t-s)g(y_s)dx_s. \quad (1.7)$$

Key task: define $\int_0^t S(t-s)g(y_s)dx_s$ for $S(\cdot) \in C^{\beta, \beta}$.

- Garrido-Atienza & Lu & Schmalz (2015)**: define rough integral using fractional calculus and solution definition by Hu & Nualart (2009).
- Hesse & Neamtu (2019)**: define rough integrals for controlled rough path $y \in C^{\beta, \beta}$ and $x \in C^\beta$ using a modified sewing lemma.
- Our idea: **A hybrid form**, i.e. define rough integral for controlled rough path using fractional calculus. Advantage: **simpler scheme with explicit Y-L typed estimate, and solution (y, y')** .

Finite dimension case: Assumptions

- $(\mathbf{H}_f)$ $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is locally Lipschitz continuous and of one-sided linear growth

$$\exists C > 0 : \langle y, f(y) \rangle \leq C(1 + \|y\|^2), \quad \forall y \in \mathbb{R}^d; \quad (1.8)$$

in addition f is of linear growth in the perpendicular direction, i.e. there exists $C_f > 0$ such that

$$\left\| f(y) - \frac{\langle f(y), y \rangle}{\|y\|^2} y \right\| \leq C_f (1 + \|y\|), \quad \forall y \neq 0; \quad (1.9)$$

- $(\mathbf{H}_g^b)$ g belongs to $C_b^3(\mathbb{R}^d, (\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max \left\{ \|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty \right\} < \infty; \quad \text{or} \quad (1.10)$$

$(\mathbf{H}_g^l)$ $g(y) = Cy$ for $C \in ((\mathbb{R}^m, \mathbb{R}^d), \mathbb{R}^d)$ such that

$$C_g := \|C\| < \infty; \quad (1.11)$$

- $(\mathbf{H}_X)$ for a given $\nu \in (\frac{1}{3}, \frac{1}{2})$, x belongs to the space $C^\nu(\mathbb{R}, \mathbb{R}^m)$ of all continuous paths which are of finite ν -Hölder norm on any interval $[s, t]$. In particular, x is a realization of a stochastic process $X_t(\omega)$ with stationary increments, such that x can be lifted into a realized component $\mathbf{x} = (x, \mathbb{X})$ of a stochastic process $(x(\omega), \mathbb{X}(\omega))$ with stationary increments, and the estimate

$$E \left(\|x_{s,t}\|^p + \|\mathbb{X}_{s,t}\|^q \right) \leq C_{T,\nu} |t - s|^{p\nu}, \quad \forall s, t \in [0, T] \quad (1.12)$$

holds for any $[0, T]$, with $p\nu \geq 1$, $q = \frac{p}{2}$ and some constant $C_{T,\nu}$.

Existence and uniqueness theorem

(H'_f) f is a locally Lipschitz continuous function which satisfies (??) and (1.9).

Theorem

- Under the assumptions (H'_f) , (H_g^b) , (H_X) , there exists a unique solution of (1.7) on any interval $[0, T]$. In addition, for each $\lambda \in (0, 1)$ small enough, there exist some generic constants $C(\lambda), \delta(\lambda)$ such that the solution satisfies the following estimates

$$\|y\|_{\infty, [0, T]} \leq e^{\delta(\lambda)T} \left(\|y_0\| + C(\lambda)T + \frac{\lambda}{2} N\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [0, T]\right) \right) =: R. \quad (1.13)$$

- Under the assumptions (H'_f) , (H_g^l) , (H_X) , there exists a unique solution of (1.7) on any interval $[0, T]$. In addition, for each $\lambda \in (0, 1)$ small enough, there exist some generic constants $C(\lambda), \delta(\lambda)$ such that the solution satisfies

$$\|y\|_{\infty, [0, T]} \leq \exp\{\delta(\lambda)T + \lambda N\left(\frac{\lambda}{16C_p C_g}, \mathbf{x}, [0, T]\right)\} \left(\|y_0\| + \frac{C(\lambda)}{\delta(\lambda)} \right) - \frac{C(\lambda)}{\delta(\lambda)} =: R. \quad (1.14)$$

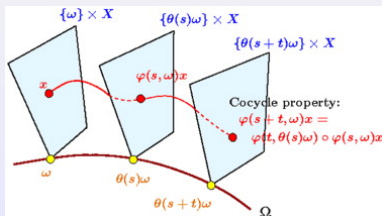
Random dynamical systems

Definition

A (continuous) random dynamical system is an NDS (θ, φ) which in addition has the following properties:

- (i), The driving system θ is an ergodic dynamical system $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t \in T})$, i.e., the base $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $(t, \omega) \mapsto \theta_t \omega$ is a measurable flow which is ergodic under $\mathbb{P}$.
- ii, The cocycle $(t, \omega, x) \mapsto \varphi(t, \omega)x$ is measurable w.r.t $\mathcal{B} \otimes \mathcal{F} \otimes \mathcal{B}(X)$ and $\mathcal{B}(X)$, respectively, where $\mathcal{B}(X)$ is the Borel σ -algebra of X , such that the family $\varphi(t, \omega, \cdot) = \varphi(t, \omega) : X \rightarrow X$ of self-mappings of X satisfies the *cocycle property*

$$\varphi(0, \omega) = id_X, \quad \varphi(t + s, \omega) = \varphi(t, \theta_s \omega) \circ \varphi(s, \omega), \quad \forall t, s \in T, \quad \forall \omega \in \Omega. \quad (2.1)$$



Generation of RDS: rough equations

Theorem

Given the measurable metric dynamical system $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ and the p -rough cocycle $\mathbf{X} : \mathbb{R} \times \Omega \rightarrow T_1^2(\mathbb{R}^m)$ as above, the system (2.2) generates a continuous random dynamical system φ over $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$, such that for any $[0, T]$ and all $\omega \in \Omega$, $\varphi(t, \omega)y_0$ is the unique solution (in the Gubinelli sense) of (1.7),

$$dy_t = f(y_t)dt + g(y_t)d\mathbf{X}_t(\omega), t \geq 0, \quad (2.2)$$

on $[0, T]$, where $\mathbf{x} = (x, \mathbb{X})$ is the projection of $\mathbf{X}_\cdot(\omega)$ on $\mathbb{R}^m \oplus (\mathbb{R}^m \otimes \mathbb{R}^m)$.

M. Hairer (2005, 2007, 2011): under fractional Brownian noises, the rough differential equation also generates a **stochastic dynamical system** $\Rightarrow$ **important to prove ergodicity!**

Random attractors

Universe: $\mathcal{D}$ a family of random sets $\hat{M} = \{M(\omega)\}_{\omega \in \Omega}$ which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$).

Definition (Crauel & Flandoli-1994)

An invariant random compact set $\hat{A} \in \mathcal{D}$ is called

(i), a pullback random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the pullback sense, i.e.

$$\lim_{t \rightarrow \infty} d(\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega) | A(\omega)) = 0; \quad (2.3)$$

(ii), a forward random attractor in $\mathcal{D}$, if $\hat{A}$ attracts any closed random set $\hat{D} \in \mathcal{D}$ in the forward sense, i.e.

$$\lim_{t \rightarrow \infty} d(\varphi(t, \omega)D(\omega) | A(\theta_t\omega)) = 0; \quad (2.4)$$

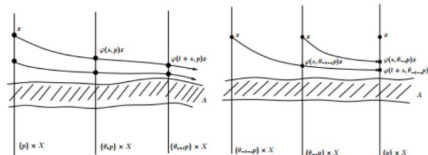


Figure 2.1: Forward attraction versus pullback attraction.

Existence of random attractor

Pullback absorbing set: $\hat{B} \in \mathcal{D}$ in a universe $\mathcal{D}$ such that $\hat{B}$ absorbs all sets in $\mathcal{D}$, i.e. for any $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega, \hat{D})$ such that

$$\varphi(t, \theta_{-t}\omega)D(\theta_{-t}\omega) \subset B(\omega), \text{ for all } t \geq t_0.$$

Theorem (Crauel, Flandoli, Schenk-Hoppe 1998: Existence and Uniqueness of Random Attractor)

Given a universe $\mathcal{D}$, assume there exists a random compact pullback absorbing set $\hat{B} \in \mathcal{D}$ which is forward invariant. Then there exists a unique random pullback attractor (which is then a weak attractor) in $\mathcal{D}$, given by

$$A(\omega) = \bigcap_{t \geq 0} \varphi(t, \theta_{-t}\omega)B(\theta_{-t}\omega).$$

Application: prove that a system has an absorbing set.

Random attractors for RDE: finite dimension

We would like to investigate the RDE of the form

$$dy_t = f(y_t)dt + g(y_t)dx_t. \quad (2.5)$$

(H'_f) and the dissipativity condition

$$\exists D_1 \geq 0, D_2 > 0 : \quad \langle y, f(y) \rangle \leq \|y\|(D_1 - D_2\|y\|), \quad \forall y \in \mathbb{R}^d, \quad (2.6)$$

(H_g) g belongs to $C_b^3(\mathbb{R}^d, (\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max \left\{ \|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty \right\} < \infty; \quad (2.7)$$

(H_X) for a given $\nu \in (\frac{1}{3}, \frac{1}{2})$, x belongs to the space $C^\nu(\mathbb{R}, \mathbb{R}^m)$ of all continuous paths which is of finite ν -Hölder norm on any interval $[s, t]$. In particular, x is a realization of a stationary stochastic process $X_t(\omega)$, such that x can be lifted into a realized component $\mathbf{x} = (x, \mathbb{X})$ of a stationary stochastic process $(x, \mathbb{X}, \cdot(\omega))$, such that the estimate

$$E \left(\|x_{s,t}\|^p + \|\mathbb{X}_{s,t}\|^q \right) \leq C_{T,\nu} |t - s|^{p\nu}, \quad \forall s, t \in [0, T] \quad (2.8)$$

holds for any $[0, T]$, with $p\nu \geq 1$, $q = \frac{p}{2}$ and some constant $C_{T,\nu}$.

Random attractors for RDE: finite dimension

$(\mathbf{H}_{\mathcal{A}})$ There exists a duration $r > 0$ and constants $D_3 > 0$ of the deterministic system such that, for any starting point $y_0 \notin \mathcal{A}$, there exists a point $\mu_0 = \mu_0(y_0) \in \mathcal{A}$ satisfying

$$\|\mu_r(y_0) - \mu_r(\mu_0)\| \leq e^{-D_3} \|y_0 - \mu_0\|. \quad (2.9)$$

Theorem (LHD-2020: finite dimension)

Assume that system (2.2) satisfies the assumptions $(\mathbf{H}'_f)$, $(\mathbf{H}_g)$, $(\mathbf{H}_X)$, then there exists a random pullback attractor $\mathcal{A}(\omega)$ such that $|\mathcal{A}(\cdot)| \in \mathcal{P}$ for any $\rho \geq 1$. If in addition $(\mathbf{H}_{\mathcal{A}})$ and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A}(\omega) \rightarrow \mathcal{A}$ (w.r.t. the Hausdorff semi-distance) as $C_g \rightarrow 0$, both in the almost sure and in $\mathcal{L}^p$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}(\omega)$ is a singleton provided that C_g is sufficiently small.

NDCong-LHD-PTHong (2020): similar results on existence of random attractors for stochastic differential delay equations. Technique: **semigroup**.

Euler scheme

For any finite partition $\Pi := \{0 = t_0 < t_1 < t_2 < \dots < t_{m-1} < t_m = T\}$ such that $|\Pi| = \sup_k |t_{k+1} - t_k|$, we consider the explicit Euler scheme of equation (1.7) to approximate the fixed solution $y(\cdot, 0, y_0)$, i.e.,

$$\begin{aligned} y_0^\Pi &= y_0; \\ y_{k+1}^\Pi &= y_k^\Pi + f(y_k^\Pi)(t_{k+1} - t_k) + g(y_k^\Pi)x_{t_k, t_{k+1}} + Dg(y_k^\Pi)g(y_k^\Pi)\mathbb{X}_{t_k, t_{k+1}}, \quad 0 \leq k \leq m-1. \end{aligned} \quad (3.1)$$

Since the solution $y(t, 0, y_0)$ is fixed on $[0, T]$, its supremum norm is bounded by a fixed number R following (1.13).

Theorem

Assume that $y(t, 0, y_0)$ is a solution of the rough differential equation (1.7) on $[0, T]$, under assumption (H_f') for f , assumption (H_g^b) or (H_g') for g , and assumption (H_X) for x . Then there exists a generic constant

$$C = C(f, g, \|x\|_{\nu, [0, T]}, T, \|y_0\|) > 0$$

for R defined in (1.13) such that for $|\Pi| < \delta$ small enough

$$\sup_{0 \leq k \leq m} \|y(t_k, 0, y_0) - y_k^\Pi\| \leq C|\Pi|^{3\nu-1}. \quad (3.2)$$

Random attractor for Euler scheme

Theorem (LHD-P.Kloeden (2021): Existence of numerical attractors)

Under the hypotheses $(H'_f), (H_g^b)$ and (H_X) , assume further that f is dissipative with (2.6) and of linear growth, i.e. there exists C_f such that

$$\|f(y)\| \leq C_f(1 + \|y\|). \quad (3.3)$$

Then there exists a $h_0 > 0$ such that for all $h < h_0$, the generated discrete-time random dynamical system φ^h from (??) admits a global random pullback attractor $A^h(\omega)$.

Theorem (LHD-P.Kloeden (2021): Upper semicontinuity of numerical attractor)

Assume $(H_g^b), (H_X)$ with the Gaussian process X and further that f is globally Lipschitz continuous and bounded, such that the dissipativity condition (2.6) is satisfied. Then φ^h admits a numerical pullback attractor $\mathcal{A}^h$ which converges to the attractor $\mathcal{A}$ a.s. in the Hausdorff semi-distance, i.e.,

$$\lim_{h \rightarrow 0} d_H(\mathcal{A}^h | \mathcal{A}) = 0 \quad \text{a.s.} \quad (3.4)$$

Thank you!

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